

## Solving Systems of Three Equations (Homework)

Solve.

Answers  
(X, Y, Z)

$$\textcircled{1} \begin{cases} 2y = x - z \\ 3 + x = y \\ 3x - 4y - 2z = -10 \end{cases}$$

$$(-10, -7, 4)$$

$$\textcircled{2} \begin{cases} 2x - 2y + 3z = 0 \\ 3x + 2y + z = -2 \\ -x - 2y + 4z = 8 \end{cases}$$

$$(-2, 1, 2)$$

$$\textcircled{3} \begin{cases} -2x + y - 2z = 7 \\ 3x = z \\ 3x - y + 2z = -8 \end{cases}$$

$$(-1, -1, -3)$$

$$\textcircled{4} \begin{cases} x = -13 - 3y + 6z \\ 2x + 3y + 2z = 2 \\ -3x + 2y - 5z = -7 \end{cases}$$

$$(-1, 0, 2)$$

$$\textcircled{5} \begin{cases} x + 3y - z = 2 \\ -x - y + 2z = 2 \\ 3x + 2y - z = 3 \end{cases}$$

$$(1, 1, 2)$$

$$\textcircled{6} \begin{cases} 2x - 2y + 5z = 7 \\ x - y + z = 2 \\ 3x + 4y - z = 2 \end{cases}$$

$$(1, 0, 1)$$

$$\textcircled{7} \begin{cases} 9x - 3y + 2z = 4 \\ 6x - 3y - 3z = -3 \\ z = 2x - y + 1 \end{cases}$$

Infinite Solutions

$$\textcircled{8} \begin{cases} 3x + 2y - 8z = -7 \\ -5x - 3y - 7z = 13 \\ 6x + 5y + 3z = -10 \end{cases}$$

$$(-5, 4, 0)$$

$$\textcircled{9} \begin{cases} 3x + y - 4z = 1 \\ 4x - y = 7 \\ x + 3y - 5z = 6 \end{cases}$$

$$(4, 9, 5)$$

$$\textcircled{10} \begin{cases} 4x - 2y + z = -10 \\ -5x + y + z = 17 \\ 10x - 2y - 2z = -30 \end{cases}$$

No Solution

\* Don't  
be afraid  
of fractions

$$\textcircled{11} \begin{cases} 6 - x = 4y + 1 + 2z \\ y = x \\ 4z = x + y \end{cases}$$

$$(5/6, 5/6, 5/12)$$